

COM668 Risks & Prototyping

Removing risk and breaking ground

Risks and prototyping

- Risks
 - Risks in software development
 - Risk management
 - Risk management plan
 - Risk monitoring and mitigation action
- Prototyping
 - What to prototype
 - How to prototype
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Risks in Software Development

- Project risk
 - Software development involves technological advancements
 - Strive for new features, efficiencies or exploiting advancements
 - Requires high levels of knowledge
 - As such software development project contains uncertainty
 - This is known as project risk.
- Risk exposure
 - Can compromise project success
 - Represent a possibility of failure
 - Risk exposure is the probability and size of the potential failure
- Risk analysis and management
 - Process of identifying and removing or reducing project risk exposure

Types of risk

- Most software projects can have four risk impact areas
 - Unproven technologies
 - User and functional requirements
 - Application and system architecture
 - Performance

Unproven technology risks

- New tools, techniques, protocols, standards, & development systems
- These increase the probability that technology risks will arise
- Improper use of new technology can lead to project failure
- Training and knowledge are critical in addressing this type of risk

User and functional requirements risks

- User needs lead to user and functional requirements
- Needs can change during a project due to knowledge discovery
- These changes can propagate throughout a project's requirements
- These disruptions can lead to failures in a poorly planned project

Application and system architecture risks

- Taking the wrong direction with system design can lead to failures
- Understanding application or system architecture is vital to success
- Knowledge will inform making sound design choices

Performance risks

- Risk management must encompass expectations on performance
- Need to ensure products are moving in the right direction

Risk Management

- Risk management means risk containment and mitigation
 - First identify risk and plan against them
 - Then be ready to act when a risk arises
 - This will help to minimise the impact to the project
- Risk management
 - Identify risks and their triggers
 - Classify and prioritise risks
 - Create a plan that links each risk to a mitigating action
 - Monitor for risk triggers during the project
 - Implement the mitigating action if any risk materialises
 - Communicate risk status throughout project

Risk Management Plan

- Catalogue risks according to type
- Create a risk management plan
- Risk management plan outlines response for each risk

RISK	LEVEL	ELIMINATION	BACKUP
Get Invited API	10/High	This risk is the first to be tackled, as it will be used for the applications main purpose of pulling in events. A prototype will be developed using the Get Invited API, thus removing any concerns about developing with it. Without this functionality the application will not be functional.	If the risk can't be eliminated, and the API becomes too complex to work with there are many alternatives. EventBrite is the most common with many online tutorials so it will be used as a suitable replacement.
Secure login	8/High	Having a secure login for the application not only provides the platform in which the app will operate but also provides user trust for the application. Apps without secure logins do not have a user base. The developer has developed many secured login systems therefore having a reference point as well as sufficient knowledge.	A basic login system can be used as a back up until a more secure login can be developed. The application can be made as a minimum viable product and therefore not require an extremely secure interface.

		project. Creating user connections has been done previously. With the chosen language of PHP not only will there be a base knowledge but a large community online of people making similar functionally.	functionality will be achieved. Allowing for extra time will insure that it is completed.
Inviting friends to events.	5/ Intermediate	Once the previous risk of being connected to friends within the application is completed the next most important feature is the ability to invite these friends to the chosen event. Tutorials online of how many people have implemented this functionality exist.	This risk needs to be developed and an alternative can't be suggested. Allowing suitable time to develop this will help eliminate the risk.
Profile Creation	4/Low	Profile creation will allow the user to personalise their experience within the application. Queries to a database, to update user information has been completed on previous projects.	If the profile creation falls out of scope with time spent on higher risks the alternative will be to provide a more basic profile system with less customisation for the user.

Risk Monitoring and Mitigation action

- Risk monitoring has to be integral to the project
- This means frequent checking during project meetings and critical events
- When a risk occurs the planned mitigation response should be taken
- Mitigating options include:
 - Accept
 - Acknowledge that a risk is impacting the project
 - Make an explicit decision to accept the risk without any changes to the project
 - Avoid
 - Adjust project scope, schedule, or constraints to minimise the effects of the risk
 - Control
 - Take action to minimise the impact or to reduce intensification of the risk
 - Monitoring
 - Monitor the project environment for potentially increasing impact of the risk
 - Only suitable for low-impact risks

Functional prototype

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What to prototype

- Pick a significant risk related to project implementation and explore it
 - Likely the biggest technical obstacle to project success
- Perhaps related to
 - Key user and functional requirements
 - Core feature or data structure (but not login)
 - New or unproven technologies
 - Technology that is new to you and which you plan to use
- Discuss the focus of your prototype with your Mentor

How to prototype

- A prototype is not a finished item
 - In this case it is focused on addressing a risk
 - It should be low fidelity, you will most likely throw it away
 - Very rarely is the appearance of significance
- Prototype from an informed position, based on your research
- Keep notes
- Allow for a few false starts and dead ends

Documenting the prototype design

- What to include depends on the element that you are prototyping
 - Small scope means a full set of design artefacts is not expected
 - Do not need to include full architecture, interface and data designs
- Items that might need to be included for the area under development
 - Use Case Diagrams
 - Activity Diagrams
 - Algorithm design
- The objective is to evidence your prototype design process

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